ModelIt!

Student Activity Pack

The Heat Island Effect

Modeling Urban Temperature Amplification, Health Impacts, and Green Infrastructure Solutions

Grade 11L2 | NGSS HS-ESS3-4, HS-ETS1-1 | Cause and Effect / Systems and System Models

STUDENT NAME _____ DATE _____

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **Why do you think a parking lot feels hotter than a park on a summer day — what's actually different about those two surfaces?**

2. **Have you ever noticed temperature differences between different neighborhoods in your city? What do you think causes those differences?**

3. **Draw a diagram showing how you think a city's buildings and roads affect temperature compared to a forest or farm.**

4. **Who do you think is most affected by extreme heat in a city — and why might some neighborhoods be hotter than others?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Urban Land Cover Composition
- Surface Heat Absorption Rate
- Waste Heat Generation
- Evapotranspiration Cooling Rate
- Ambient Air Temperature
- Heat-Related Health Impact

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Current Urban Baseline

60-80% impervious surface, minimal tree canopy in underserved areas

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Climate Change Amplification

Current heat island plus 3-5°F baseline warming

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Green Infrastructure Intervention

30% tree canopy, 50% cool roofs, connected green corridors

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Air Quality Degradation — Higher urban temperatures accelerate the chemical formation of ground-level ozone (smog) and increase particulate matter suspension, creating a heat-air quality feedback loop where the hottest neighborhoods also have the worst air quality — compounding respiratory health impacts

Energy Demand Feedback — Each 1°F increase in urban temperature increases peak electricity demand by 1.5-2% for air conditioning, stressing power grids, increasing fossil fuel combustion at power plants, and generating more waste heat — a positive feedback loop where cooling demand amplifies the heat that drives it

Stormwater Runoff Intensity — Impervious urban surfaces that cause heat islands also prevent water infiltration, increasing stormwater runoff volume and velocity during rain events — green infrastructure that addresses heat (permeable pavement, rain gardens, tree canopy) simultaneously reduces flood risk, creating co-benefits

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

Why are cities dramatically hotter than the surrounding countryside — and who pays the price when urban heat becomes deadly? Use evidence from your simulation to support your answers.

1. **Why does your model show that the urban heat island is not just a climate problem but an environmental justice problem — and how do historical decisions about where to invest in green space still determine who suffers most from heat today?**

2. **How does the air conditioning feedback loop in your model reveal a paradox where individual cooling solutions make the collective problem worse?**

3. **What does your model reveal about why nighttime heat is more dangerous than daytime heat for human health, and how urban materials prevent nighttime cooling?**

4. **How do the co-benefits of green infrastructure — cooling, air quality, stormwater, mental health, property value — change the cost-benefit calculation compared to analyzing heat reduction alone?**

5. **What does your model suggest about the tension between designing cooling solutions for the hottest neighborhoods and the risk that those improvements trigger gentrification and displace the people they were meant to protect?**

8 STEM Challenge: Design an Urban Heat Equity Action Plan

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YOUR ENGINEERING MISSION

Design a neighborhood-scale cooling strategy that reduces peak summer temperatures by at least 8°F in the hottest, most vulnerable census tract of a major city within 15 years, using a combination of green infrastructure, cool surfaces, and community design.

THE SCENARIO

A city health department has identified a census tract where heat-related emergency room visits are 4x the city average. The neighborhood has 5% tree canopy (versus 25% citywide), 85% impervious surface cover, no parks within walking distance, and a population that is 70% low-income with high rates of outdoor workers and elderly residents without air conditioning. Your team has been funded to design a comprehensive cooling plan.

Engineering Design Thinking

1. WHAT TEMPERATURE MONITORING DATA WOULD YOU NEED TO MAP HEAT EXPOSURE AT THE BLOCK LEVEL AND IDENTIFY THE HOTTEST MICROCLIMATES WITHIN THE NEIGHBORHOOD?

2. HOW WOULD YOU PRIORITIZE COOLING INVESTMENTS — SHOULD YOU START WITH THE HOTTEST BLOCKS, THE BLOCKS WITH THE MOST VULNERABLE RESIDENTS, OR THE BLOCKS WHERE COOLING INFRASTRUCTURE IS MOST COST-EFFECTIVE?

3. WHAT COMBINATION OF SHORT-TERM SOLUTIONS (SHADE STRUCTURES, COOL ROOFS, MISTING STATIONS) AND LONG-TERM SOLUTIONS (TREE PLANTING, PARK CREATION, GREEN CORRIDORS) WOULD YOUR PLAN INCLUDE?

4. HOW WOULD YOU ENGAGE COMMUNITY RESIDENTS IN THE DESIGN PROCESS TO ENSURE THE PLAN REFLECTS THEIR PRIORITIES AND AVOIDS UNINTENDED DISPLACEMENT?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! Explain your design using scientific evidence from the simulation:

Real Career Connection

Urban Climate Scientists study how cities interact with weather and climate systems, using sensor networks, remote sensing, and computational modeling to map heat vulnerability and design cooling strategies. They work for city planning departments, public health agencies, and environmental consulting firms, earning $70,000–$130,000/year. Environmental Justice Planners who design equitable climate adaptation strategies for underserved communities earn $60,000–$120,000/year at government agencies and nonprofit organizations.